Exp 12 End - End communication at Transport layer

a Implement echo client server using TCP/UDP Socket

TCP

```
import Socket
import threading

def start_server():
    Server = socket.Socket (Socket.AF_INET,
Socket.SOCK_STREAM)
    Server.bind(('127.0.0.1', 5000))
    server.listen(1)
    Print ("[Server] Listening on 127.0.0.1:50
    conn, addr = server.accept()
    Print (f "[server] connected by{addr}")
    while True:
        data = conn.recv(1024).decode()
        if not data or data.lower == 'quit':
            Print ("[sever] connection close
by client.")
            break
```

```
        Print(f"[Server] Received: {data}")
        conn.send(f"Server received: {data}".encode())
    conn.close()
server.close()

def start_client():
    client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    client.connect(('127.0.0.1', 5000))
    Print("[client] connected to server")

    while True:
        msg = input("Enter message")
        client.send(msg.encode())
        if msg.lower() == 'quit':
            break
        response = client.recv(1024).decode()
        Print(f"[server Response] {response}")
    client.close()
```

```
if __name__ == "__main__":
    server_thread = threading.Thread
(target = start_server, daemon = True)
    server_thread.start()

    start_client()
```

Output :

```
hello
Server : hello
Server Response : hello
```

```
Hi
Server Respone : Hi
connection closed.
```

UDP

```
import Socket
import threading
def udp_server():
    s = Socket.Socket(socket.AF_I
        Socket.SOCK_DGRAM)
```

```
    s.bind(('127.0.0.1', 12000))
    Print("[server] on 127.0.0.1:12000")
    while True:
        data,addr = s.recvfrom(1024)
        msg = data.decode()
        if msg.lower()='quit':
            Print(f"[server] client {addr}
ended the connection.")
            break
        Print(f"[server] Received from
{addr}:{msg})
        s.Sendto(data,addr)
    S.close()
    Print("[server] stopped")

def udp-client():
    c = socket.socket(socket.AF_INET,
Socket.SOCK_DGRAM)
    server = ('127.0.0.1', 12000)
```

```
while True:
    msg = input("Enter message : ")
    c.sendto(msg.encode(), server)
    if msg.lower() == 'quit':
        Print("[client] closing connection ....") break
    data, _ = c.recvfrom(1024)
    Print("Echo from Server :", data.decode())
c.close()
Print("[CLIENT] Stopped.")

if __name__ == "__main__":
    threading.Thread(target=udp_s
daemon = True).start() udp_client()
```

output-

```
Echo from Server : hi
Enter message : hello
Received : hello
Enter message : how are you
Echo from server : howare you
```

Result :

Therefore, The implement echo client/server TCP/IP Socket had executed Successfully